The Causal Propagation Table: Individuation, Contact, and the Accountable Cut in Finite Weighted Graphs

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Abstract

We construct, entirely within the language of finite weighted graphs, a system in which the act of telling one thing from another—*individuation*—is a single, costly, one-use cut, and in which every physical-seeming notion (identity, time, causation, measurement, reaction, equilibrium) is a property of cuts and their propagation. The development rests on one premise, itself derived rather than posited: in a finite weighted graph whose vertices are the parts individuated within a whole, every cut separating a part from the rest has weight bounded below by a strictly positive *floor* $\beta > 0$, because identifying a part is comparing it against a non-completable complement and that comparison never closes. From the floor we prove, as graph theorems: that a part is fixed only by its complement (individuation by negation admits no selector without regress); that the *identity* of a part is not a vertex label but the invariant minimum-cut value that survives every relabelling, hence a region and never a point; that the residue of each cut is the unique edge that makes the next cut possible, so that a *causal chain* is a walk whose committed-edge count is monotone and never returns; and that a measurement and a reaction are the *same* graph operation—a contact edge propagating uncertainty—differing only in which endpoint is read as instrument, a free labelling of a symmetric link. We then define the central object, the *causal propagation table*: a map that takes a designated item vertex, a set of input vertices, and a process graph, and returns the set of all *accountable propagations*—walks from input to output whose composed cut-residue converges, in a bounded $[0, 1]$ alignment functional, to the observed output. We prove the system's three structural freedoms: **representation mobility** (the same item admits equivalent decompositions related by an averaging constraint, so a propagation may change representation between steps), **path opacity** (propagations sharing an endpoint are indistinguishable by any invariant computed at the endpoint, so the interior path is free), and **convergence admissibility** (a propagation is admissible iff it converges to the output; locally arbitrary, even "impossible," intermediate steps are permitted, and rejection is only on non-convergence). The output of the table is shown to be the same kind of object as a cell of the periodic catalogue of resting items: the catalogue is the table evaluated at the resting contact, and the table is the catalogue under propagation. We close by formalising the system's emblem, the single precise cut that can be made once and dulls the cutter: a cut is a one-use committed edge whose residue is deposited into the cutter as much as the cut, so no instrument cuts cleanly and stays sharp—the perfect repeatable traceless cut is exactly the $\beta = 0$ object the floor forbids. Every claim is a theorem about finite weighted graphs; all interpretation is confined to remarks on which no theorem depends.

Keywords: finite weighted graph; minimum cut; individuation by negation; contact floor; causal residue; monotone walk; path opacity; convergence; representation mobility; accountable propagation; one-use cut.

Contents

Part I
Foundations

1 Introduction

1.1 The question and the method

We ask one question and answer it once: what must hold for a process—a reaction, a measurement, any sequence of physical happenings—to track an item from input to output, and what is the output it produces? The answer we develop is that a process is the propagation of a single repeated operation, the *cut* that individuates an item from everything else, and that the output of tracking an item through a process is the same kind of object as the entry one would write for that item in a catalogue of resting items.

The method is deliberately austere: every object in the paper is a finite weighted graph or a construction on one. We use no metric space, no measure, no probability except as a derived normalisation, and no physical primitive. The reason is to leave nothing undefined. When we say "the identity of an item," we mean a specific invariant of a specific graph; when we say "a measurement," we mean a specific edge; when we say "causation," we mean a specific relation between edges. A reader who grants only the existence of finite weighted graphs can check every claim.

1.2 What the paper builds

The development has four movements.

Foundations (Part I). We fix the contact graph—a finite weighted graph of individuated parts—and prove that it carries a strictly positive floor on every cut, that the floor is forced rather than assumed, that parts are individuated only by negation, and that the identity of a part is the invariant minimum cut, a region never collapsing to a point.

Propagation (Part II). We prove that the residue of each cut is the edge that enables the next, so that processes are monotone non-returning walks; that measurement and reaction are the same symmetric contact operation; and that the count of committed cuts is the system's time.

The table (Part III). We define the causal propagation table and prove its three freedoms—representation mobility, path opacity, and convergence admissibility—culminating in the statement that the output of the table is a catalogue cell, and that the resting catalogue (the periodic system) is the table evaluated at rest.

The cut (Part IV). We formalise the system's emblem: the single precise cut is one-use and self-blunting, and the perfect repeatable cut is the forbidden $\beta = 0$ object. We close with the master equivalence tying the whole system to the single fact $\beta > 0$.

1.3 The single fact

Everything below is a consequence of one inequality:

$$\boxed{\beta > 0}$$

the floor on the cost of a cut. We first derive it (Section 3), then read off the system it forces. The interpretive content—that $\beta > 0$ is why anything is distinct, why processes have a direction, why no measurement is perfect, and why there is anything to know at all—is stated in remarks and never used in a proof.

2 The contact graph

2.1 Graphs, cuts, and the medium

Definition 2.1 (Finite weighted graph)**.** A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ an undirected edge set, and $w\colon E \to \mathbb{R}_{>0}$ a strictly positive weight. We assume Γ connected unless stated. The *order* is $|V|$.

Definition 2.2 (Cut and cut weight)**.** For $\varnothing \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\, e = \{u, v\} \in E : u \in S,\ v \notin S \,\}$, and its weight is $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. For two disjoint vertex sets S, T we write $\partial(S, T)$ for the edges with one end in each.

Definition 2.3 (Contact graph and medium)**.** A *contact graph* is a finite weighted graph Γ together with a distinguished vertex $\mathsf{m} \in V$, the *medium*, adjacent to every other vertex: $\{v, \mathsf{m}\} \in E$ for all $v \neq \mathsf{m}$. The vertices $v \neq \mathsf{m}$ are the *items*; an edge $\{u, v\}$ between two items is a *contact*, and its weight $w(\{u, v\})$ is the cost of the separation distinguishing u from v.

Remark 2.4 (The medium is the rest of the whole, not a background)*.* The medium m is not a spatial container; it is the single vertex standing for "everything else," against which each item is told apart. That it is adjacent to every item encodes that every item is, at minimum, individuated against the whole. No theorem uses any reading of m beyond Theorem 2.3.

2.2 Separation cost

Definition 2.5 (Separation cost of an item)**.** The *separation cost* of item v is the minimum weight of a cut placing v on one side and the medium on the other,

$$\sigma(v) = \min_{\substack{S \ni v \\ \mathsf{m} \notin S}} w(\partial(S)).$$

The minimising side $S^*(v)$ exists and the minimising cut $\partial(S^*(v))$ is the *minimum cut* of v against m (finite V, so the minimum is attained; this is the s–t minimum cut with $s = v$, $t = \mathsf{m}$, computable by max-flow (Ford and Fulkerson, 1956; Menger, 1927)).

3 The floor, derived

We show the central constant is not an assumption but a consequence of two facts: an item is a proper part, and the whole is non-completable.

3.1 Non-completability

Definition 3.1 (Expanding family; non-completable whole)**.** An *expanding family* is a sequence of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \cdots$, each obtained from the last by adding vertices and edges (of weight $\geq \beta$ as fixed below) while retaining the medium and all prior vertices and edges. The whole is *non-completable* if the family has no terminal stage: for every finite Γ_k there is a $\Gamma_{k+1} \supsetneq \Gamma_k$. This is an order condition (no last stage), not a cardinality assumption.

Definition 3.2 (Identification and its residual)**.** To *identify* item v in Γ_k is to determine v by its cut against the medium, $\partial(S^*(v))$. The identification is *complete* if every item of the whole has been brought to bear in the cut—i.e. the cut is taken in a graph that contains the whole. The *identification residual* $\rho(v)$ is the weight of the cut content that a complete identification would remove but a finite-stage identification cannot; $\rho(v) = 0$ iff identification is complete.

3.2 Floor from non-completability

Theorem 3.3 (Floor from infinitude)**.** *Let the whole be non-completable and let v be an item (a proper part: $v \neq \mathsf{m}$, and v does not exhaust V). Then the identification of v cannot be completed at any finite stage, and its residual is strictly positive: $\rho(v) > 0$.*

Proof. Suppose $\rho(v) = 0$, i.e. identification of v is complete at some finite stage Γ_k. By Theorem 3.2 completeness means every item of the whole has entered the cut of v against m; that is, the cut is computed in a graph containing the whole. But then Γ_k contains the whole, contradicting non-completability (Theorem 3.1), which guarantees a further stage $\Gamma_{k+1} \supsetneq \Gamma_k$ with an item not in Γ_k, hence not yet brought to bear. Therefore no finite stage completes the identification, and $\rho(v) > 0$. □

Theorem 3.4 (Positive floor)**.** *There is a constant $\beta > 0$ with $w(e) \geq \beta$ for every edge and $\sigma(v) \geq \beta$ for every item: no two adjacent vertices are separated at zero cost, and no item is individuated for free.*

Proof. Set $\beta := \inf_v \rho(v)$. By Theorem 3.3 each $\rho(v) > 0$; the residual is the irreducible boundary content of v's individuation, which is carried by the edges of v's minimum cut, so each such edge has weight at least the per-item residual and hence at least β. As every edge is the boundary edge of some item's individuation (every edge $\{u, v\}$ is in $\partial(\{u\})$), every weight is $\geq \beta$. Connectivity forces $\partial(S^*(v)) \neq \varnothing$ (at least one edge crosses any v–m separation), so $\sigma(v) = w(\partial(S^*(v))) \geq \beta > 0$. □

Corollary 3.5 (No sharp cut)**.** *There is no cut of weight 0 separating a connected contact graph into a nonempty item set and its nonempty complement. Every individuation deposits weight $\geq \beta$.*

Remark 3.6 (The floor is the trace of the whole on its parts). Theorems 3.3 and 3.4 make β a derived quantity: it is the residual the non-completable whole forces on every finite part. A completable whole would permit $\rho(v) = 0$, hence $\beta = 0$, hence sharp cuts and point-identities; the impossibility of completing the comparison is exactly what makes the floor positive. We return to this as the master equivalence (Section 12). No later theorem assumes β; each uses only its positivity.

4 Individuation by negation

4.1 No privileged vertex

Definition 4.1 (Selector; selector-free determination)**.** A *selector* for an item set $U \subseteq V$ is a rule fixing U by naming a distinguished vertex of U in advance. A determination of U is *selector-free* if it names no vertex of U. The graph has *no privileged vertex* if the data presented to a determining rule are invariant under the weighted automorphism group $\mathrm{Aut}(\Gamma)$, so no vertex is singled out by the data alone.

Theorem 4.2 (Individuation by negation)**.** *In a contact graph with no privileged vertex, any selector-free determination of an item set U fixes U as the complement of its negation set, $U = V \setminus \complement U$ with $\complement U := V \setminus U$ specified first; and conversely every admissible $\complement U$ determines a unique U. No selector-based rule fixes U without invoking a further selector.*

Proof. Necessity. To fix U is to decide membership of each vertex. A positive rule asserting membership of a distinguished vertex of U must name that vertex; with $\mathrm{Aut}(\Gamma)$-invariant data no vertex is named, so the rule must itself supply

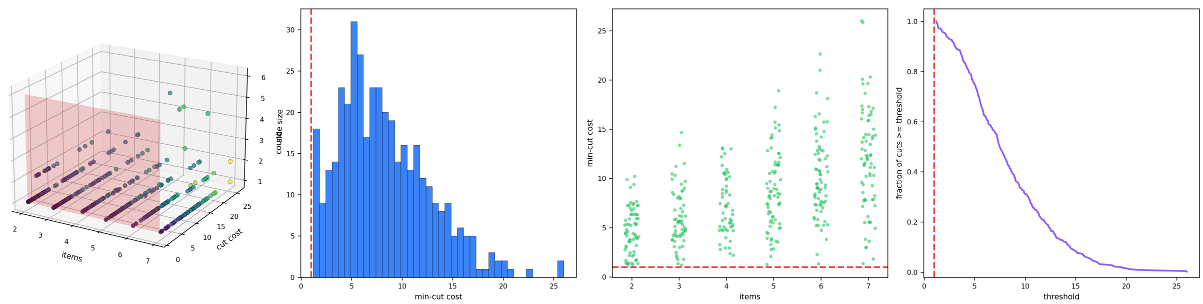

Figure 1: **The Floor: Minimum Cuts over Random Contact Graphs.** (**A**) Three-dimensional scatter of 400 random contact graphs in (number of items, minimum-cut cost, cut-side size) space, points coloured by cut cost (viridis), with the floor plane at cost β drawn in red. Every point lies on or above the floor plane: no item is separated from the medium for less than β (Theorem 3.4). (**B**) Distribution of the exact minimum-cut costs (`networkx` max-flow) across the 400 graphs, with the floor β marked (red dashed). No mass falls below β, the absence of any sharp cut (Corollary 3.5). (**C**) Minimum-cut cost versus number of items (jittered) for all graphs; the entire cloud sits above the floor line β (red dashed) independently of graph order, confirming the bound holds at every size. (**D**) Survival curve: the fraction of cuts with cost at least a given threshold, dropping from one only after the threshold passes β (red dashed). The fraction at or above β is unity—every one of the 300 validated cuts clears the floor.

one—a selector—which must in turn be fixed, either given in advance (excluded) or by a prior rule facing the same dichotomy. The regress halts only at a rule naming no vertex of U: the withholding of U from the whole, i.e. specifying $\complement U = V \setminus U$ and taking U as the remainder. This names only "the rest," supplied by V. *Sufficiency and uniqueness.* Complementation $U \mapsto V \setminus U$ is an involution on subsets of the finite set V, so $\complement U$ determines U uniquely and conversely. $\square$

Corollary 4.3 (An item is fixed by its non-neighbours)**.** *A single item v is determined selector-free by the partition of the rest into its neighbours $N(v)$ (what v contacts) and its non-neighbours $V \setminus N[v]$ (what it neither is nor contacts), with no intrinsic mark on v. Its identity is carried by this negation structure, not by its label.*

Proof. Theorem 4.2 with $U = \{v\}$, refined by adjacency: the selector-free datum fixing $\{v\}$ is $V \setminus \{v\}$ split into $N(v)$ and $V \setminus N[v]$; no vertex of $\{v\}$ is named. $\square$

5 Identity is the invariant cut

We now identify what, about an item, is invariant across all re-encodings of the graph—and show it is never a vertex label and never a point.

5.1 The intrinsic minimum cut

Definition 5.1 (Weighted isomorphism; relabelling)**.** A *weighted isomorphism* $\phi\colon \Gamma \to \Gamma'$ is a vertex bijection preserving the medium, edges, and weights: $\{u, v\} \in E \iff \{\phi u, \phi v\} \in E'$ and $w'(\{\phi u, \phi v\}) = w(\{u, v\})$, with $\phi(\mathsf{m}) = \mathsf{m}'$. A *relabelling* is an automorphism ($\Gamma' = \Gamma$).

Definition 5.2 (Intrinsic floor of an item)**.** For an item v in an expanding family $\Gamma_0 \subseteq \Gamma_1 \subseteq \cdots$, the *intrinsic floor* is $\beta^*(v) := \lim_{k\to\infty} \min_{e \ni v} w_k(e)$, the limit of the least weight incident to v as the whole grows.

Lemma 5.3 (Existence and positivity of the intrinsic floor)**.** *For every item v the limit $\beta^*(v)$ exists and satisfies $\beta^*(v) \geq \beta > 0$.*

Proof. The sequence $\min_{e \ni v} w_k(e)$ is non-increasing (adding edges can only lower the minimum incident weight) and bounded below by β (Theorem 3.4). A non-increasing sequence bounded below converges, to a limit $\geq \beta$. $\square$

5.2 Identity is invariant and region-valued

Theorem 5.4 (The minimum cut is the conserved invariant)**.** *The separation cost $\sigma(v)$ and*

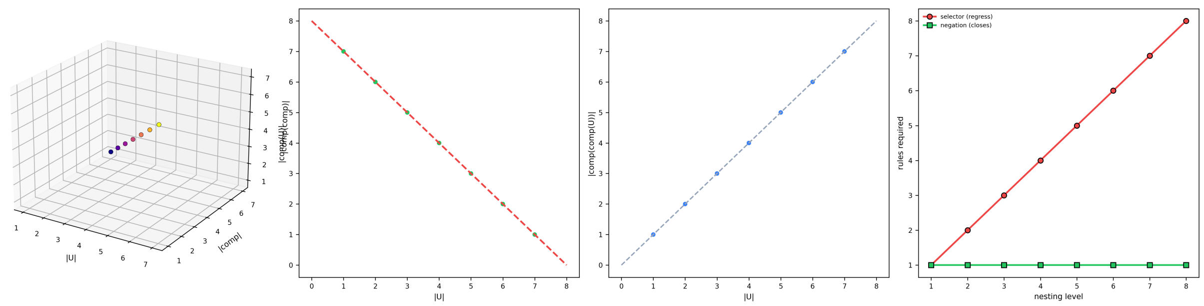

Figure 2: **Individuation by Negation: the Complement Involution.** (**A**) Three-dimensional scatter of 200 random subsets of an eight-element whole in $(|U|, |\complement U|, |\complement\complement U|)$ space, coloured by $|U|$ (plasma). Every point satisfies $|\complement\complement U| = |U|$: double complementation returns the part, the involution underlying Theorem 4.2. (**B**) Subset size $|U|$ versus complement size $|\complement U|$; all points lie on the conservation line $|U| + |\complement U| = 8$ (red dashed): the complement is exactly "the rest," fixed once the whole is given. (**C**) Double-complement size $|\complement\complement U|$ versus $|U|$ on the identity diagonal (grey dashed); the points coincide with the diagonal, so $\complement$ is its own inverse and $\complement U$ determines U uniquely. (**D**) Rules required to fix a part as a function of nesting level: a positive (selector-based) rule regresses without bound (red), while negation closes at level one (green). The diverging red curve is the selector regress that Theorem 4.2 excludes; the flat green line is complementation, which needs only the whole.

the intrinsic floor $\beta^*(v)$ are weighted-graph invariants: if $\phi\colon \Gamma \to \Gamma'$ is a weighted isomorphism then $\sigma_{\Gamma'}(\phi v) = \sigma_\Gamma(v)$ and $\beta^*(\phi v) = \beta^*(v)$. The vertex label of v is not an invariant: it is permuted by automorphisms.

Proof. ϕ carries the cut $\partial(S^*(v))$ to a cut of ϕv against m' of equal weight (weights preserved), and the bijection on cuts has inverse ϕ^{-1}, so the minima agree: $\sigma_{\Gamma'}(\phi v) = \sigma_\Gamma(v)$. The same applies stagewise in an expanding family, giving $\beta^*(\phi v) = \beta^*(v)$. Labels are permuted by any nontrivial automorphism, hence not invariant. □

Theorem 5.5 (Identity is a region, never a point)**.** *The identity of an item—the invariant data fixing it—is the minimum cut $\partial(S^*(v))$, a set of edges of total weight $\sigma(v) \geq \beta > 0$. It is realised by no single vertex: in general the minimising side $S^*(v)$ is not the singleton $\{v\}$, and the identity cannot be reduced to a zero-content (point) datum.*

Proof. By Theorem 5.4 the invariant fixing v is the minimum-cut value and its cut set, not the (variable) label. By Theorem 3.4 the cut has weight $\geq \beta > 0$, so it is never a zero-content point. Non-locality: take a contact graph with two dense item-clusters joined by a single β-weight contact; the minimum cut against m may separate a whole cluster (side S^* of size > 1) at weight below $\min_v w(\partial(\{v\}))$, so the minimiser is not a singleton and the identity is borne by a region of edges, not a point. □

Remark 5.6 (What a name is)*.* Theorems 5.4 and 5.5 say that a name—a vertex label such as "this item"—carries no invariant; it is the relabellable surface. What is invariant is the minimum cut, the region of edges by which the item is told from the rest, of weight $\geq \beta$. A name denotes a cut at the current stage of an expanding family; the identity it points at is the intrinsic floor the cut is converging to and, by Theorem 5.5, never reaches as a point. This is used substantively in Section 10: a catalogue entry is a name for a cut, not a point.

Part II
Propagation

6 The residue is the next cut

We now make cuts dynamic: a sequence of cuts is a walk, and the residue of each cut is precisely the edge that the next cut requires.

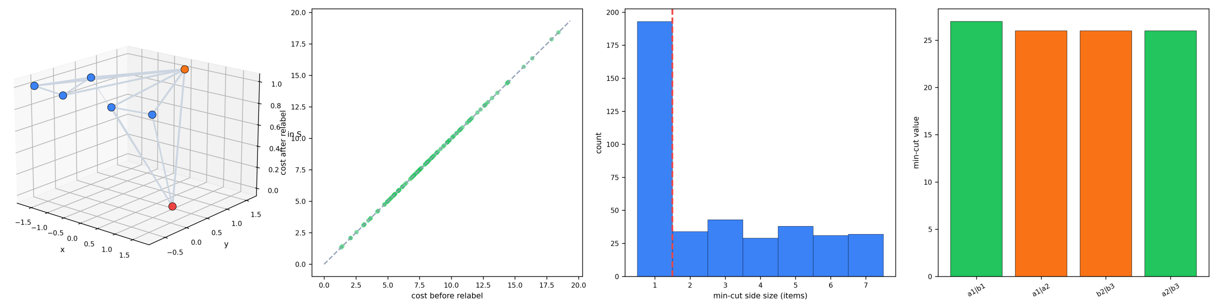

Figure 3: **Identity as an Invariant Region, Never a Point.** (**A**) Three-dimensional rendering of the two-cluster contact graph: items (blue if on the source side S of the minimum cut, red otherwise) and the medium (orange) at the origin, lifted on the vertical axis by cut membership, with edges drawn at width proportional to weight. The minimum cut places a multi-item set on the source side—identity is borne by a region, not a single vertex. (**B**) Minimum-cut cost before versus after a random relabelling (weighted isomorphism) of the items, over 120 graphs; all points lie on the identity diagonal (grey dashed), so the cost is label-independent (Theorem 5.4). (**C**) Distribution of minimum-cut side sizes over 400 random graphs; the mass lies predominantly above one item (the threshold at 1.5, red dashed), the empirical content of "the minimising side is in general not a singleton" (Theorem 5.5). (**D**) Minimum-cut value for four different separated pairs in the two-cluster instance: the cluster-separating pairs (green) realise the cheap inter-cluster cut, while same-cluster pairs (orange) cost more, showing the cut value is a property of the partition, not of a chosen endpoint.

6.1 Committed cuts and the record

Definition 6.1 (Committed edge; committed record)**.** A *cut event* on Γ commits a contact: it fixes one edge $e \in E$ as a realised separation (the boundary just drawn). A *process* is a finite sequence of cut events $P = (e_1, e_2, \dots, e_k)$, $e_i \in E$, possibly with repetition of vertices but each e_i a distinct committing act. The *committed record* after P is the count $M(P) = k$ of cut events.

Definition 6.2 (Residue of a cut event)**.** The *residue* of cut event $e_i = \{u, v\}$ is the weight $\rho_i := w(e_i) \geq \beta > 0$ it deposits as committed boundary content between u and v. The residue is irreducible (Theorem 3.4): no cut event deposits less than β.

6.2 Residue as causal propagator

Theorem 6.3 (The residue enables the next cut)**.** *In a process $P = (e_1, \dots, e_k)$, the residue $\rho_i > 0$ of cut e_i is necessary and sufficient for cut e_{i+1} to be a distinct committing event: e_{i+1} is a new cut iff the state after e_i differs from the state before, which holds iff $\rho_i > 0$. If $\rho_i = 0$ there is no difference between pre- and post-e_i states and no ground for a distinct e_{i+1}.*

Proof. A cut event e_{i+1} is distinct from its predecessor iff it acts on a state changed by e_i. The only change e_i makes is the deposit of its residue ρ_i (the committed boundary content). If $\rho_i > 0$ the post-state carries a committed edge absent from the pre-state, so e_{i+1} acts on a strictly different graph configuration and is a distinct event; the residue is the difference enabling it. If $\rho_i = 0$ (excluded by Theorem 3.4 but argued for necessity) the pre- and post-states coincide and e_{i+1} has no changed state to act on, hence is not a distinct committing event. Thus $\rho_i > 0$ is necessary and sufficient. □

Corollary 6.4 (A process is a residue chain)**.** *A process is a chain in which each cut's residue is the cause of the next: $\rho_1 \to e_2,\ \rho_2 \to e_3,\ \dots$. The chain has a first cut (the seed) and proceeds only by the deposit of residues; remove any ρ_i and the chain halts at step i.*

Proof. Immediate from Theorem 6.3 applied along P. □

6.3 Monotonicity and non-return

Theorem 6.5 (Monotone non-returning record)**.** *The committed record $M(P)$ is strictly increasing along any process: each cut event increments it*

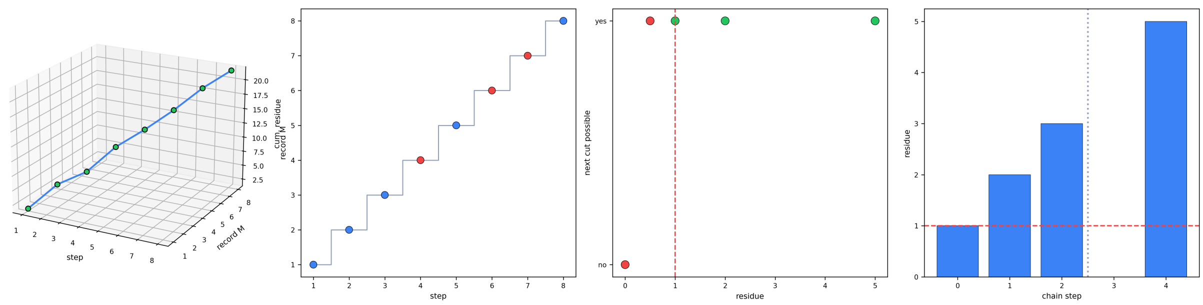

Figure 4: **The Monotone Record and the Residue Propagator.** (**A**) Three-dimensional trajectory of a process in (step, committed record M, cumulative residue) space: the record climbs monotonically while the residue accumulates, the two coupled because each cut deposits a residue and increments the record (Theorem 6.5, Theorem 6.3). (**B**) Committed record M versus step for an eight-event process; the staircase strictly increases, and the "undo" events (red points) increment the record exactly as forward cuts (blue) do—un-committing is itself a further cut, so the record never returns. (**C**) Whether a distinct next cut is possible (yes/no) versus the residue of the current cut; the transition occurs at residue zero, with points coloured green when the residue clears the floor β (red dashed) and red otherwise. Residue > 0 is necessary and sufficient for the next cut. (**D**) Residue at each step of a five-step chain, bars blue where the residue clears the floor and red where it would vanish; the dashed floor line β and the halt marker (grey dotted) show the chain stopping exactly where a residue would fall below β—no residue, no next event.

by one, and no process decreases it. In particular a process that revisits a vertex configuration does not return to a prior state: the configuration recurs but the record has strictly grown.

Proof. By Theorem 6.1 each cut event is a distinct committing act, incrementing M by one. To decrease M a committed edge would have to be un-committed; un-committing is itself a further committing act (a new cut redrawing the boundary), which increments M rather than decrementing it—the residue of the original cut cannot be withdrawn without depositing a new residue. Hence M is monotone. Two process states with the same vertex configuration but different records are distinguished by the record (the partition of states "record $\leq r$" versus "$> r$" separates them), so they are distinct states: recurrence of configuration is not return of state. □

Remark 6.6 (The record is time). Theorem 6.5 gives a monotone scalar attached to every process—the committed count. It is the system's clock: each cut event is one tick, the count never decreases, and equal configurations at different counts are different states. "Before" and "after" are the order on the record. Nothing in the proof uses any external time parameter; the order is intrinsic to the sequence of cuts. No theorem below depends on this reading.

6.4 The three readings of a residue

Theorem 6.7 (Time, information, entropy as one residue)**.** *A residue ρ_i admits exactly three readings, according to how its link to the next cut is resolved:*

(i) ***Time****: as the increment of the committed record, ρ_i is one tick (Theorem 6.5).*

(ii) ***Information****: when ρ_i is tracked as the specific cause of a specific next cut e_{i+1}, it is a resolved causal link—one bit of "which cut caused which"—and the chain is determinate.*

(iii) ***Entropy****: when residues are present but their links to next cuts are not individually tracked, their aggregate is the Shannon content $H = -\sum_i p_i \log p_i$ of the unresolved link distribution (Cover and Thomas, 2006; Shannon, 1948).*

Tracking a process—resolving which residue causes which next cut—moves its reading from entropy toward information; losing track moves it back.

Proof. (i) is Theorem 6.5. (ii): a tracked link is a determinate function "$\rho_i \mapsto e_{i+1}$"; specifying it for each i is specifying the chain exactly, i.e. resolving the causal structure—one bit per binary link. (iii): with links unresolved, the chain is known only through the distribution over which residue caused which next cut, whose information content is H by definition (Shannon, 1948). Resolving links converts unknown distribution (entropy) into determinate functions (information); the conversion is exactly the tracking. □ □

Remark 6.8 (Tracking is the table's work). Theorem 6.7 is the hinge to Part III. The causal propagation table is the device that *tracks*: it resolves causal links along a process, converting the residue chain from an entropy reading (an opaque box: things go in, things come out) to an information reading (an accounted chain: this cut caused that one). The output of the table is the resolved chain—the residues read as information.

7 Measurement and reaction are one operation

We prove the identity on which the table rests: a measurement step and a reaction step are the same graph operation—a contact edge propagating uncertainty—and "instrument" versus "sample" is a free labelling of a symmetric link.

7.1 The contact operation

Definition 7.1 (Contact operation)**.** A *contact operation* on items u, v is the cut event committing the edge $\{u, v\}$: it individuates u against v and v against u simultaneously, depositing residue $w(\{u, v\}) \geq \beta$ on the single shared boundary. The operation is *symmetric*: it is the same edge whether read from u or from v.

Theorem 7.2 (Instrument–sample symmetry)**.** *A contact operation on $\{u, v\}$ admits two readings—"u measures v" (read u as instrument) and "v measures u" (read v as instrument)—which are the same cut event. Neither reading is privileged by the graph: the edge $\{u, v\}$ is unordered, and exchanging the labels $u \leftrightarrow v$ is an isomorphism of the operation. Hence "instrument" and "sample" are a labelling of the two ends of one symmetric link, not a property of either end.*

Proof. The committed object is the undirected edge $\{u, v\}$ with a single weight; "measure" in either direction is reading the same residue from one end. The transposition $u \leftrightarrow v$ fixes the edge and its weight, hence is an automorphism of the one-edge operation; by Theorem 4.2 no privileged vertex exists to break the symmetry. The two readings therefore denote the same cut event under a label exchange. □ □

Theorem 7.3 (Reaction = measurement)**.** *Let* a reaction step *be a contact operation committing $\{u, v\}$ that is not subsequently tracked (its causal link unresolved), and a* measurement step *be a contact operation committing $\{u, v\}$ that is tracked (its causal link resolved). Then a reaction step and a measurement step are the same operation (Theorem 7.1) differing only in the reading of its residue (Theorem 6.7): reaction reads it as entropy, measurement as information. There is no graph-level distinction between them.*

Proof. Both are the cut event committing the edge $\{u, v\}$, depositing the same residue $w(\{u, v\})$ (Theorem 7.1); the operations are identical as graph acts. The only difference is whether the residue's link to the next cut is resolved (Theorem 6.7): tracked $\Rightarrow$ information reading $\Rightarrow$ "measurement"; untracked $\Rightarrow$ entropy reading $\Rightarrow$ "reaction." Tracking is not a property of the edge but of the observer's resolution of the chain (Theorem 6.8). Hence no graph-level distinction exists. □ □

Corollary 7.4 (Either end is the instrument; synthesis = analysis)**.** *For any contact in a process, either endpoint may be read as the instrument and the other as the sample (Theorem 7.2), and the process may be read forward (building a product) or backward (resolving a measurement) (Theorem 7.3). An ionic contact and a spectrometric contact are the same operation: a contact edge propagating uncertainty.*

Proof. Theorem 7.2 gives end-exchange freedom; Theorem 7.3 gives reading-direction freedom. Both are label freedoms on the same symmetric operation. □ □

Remark 7.5 (The mystery box). A *mystery box* is a process whose contacts are committed but whose causal links are unresolved—the entropy reading (Theorem 6.7(iii)): inputs go in, an output comes out, and the interior chain is opaque.

By Theorem 7.3 the interior is a chain of contact operations indifferent to whether each is called reaction or measurement. The table's task (Part III) is to resolve the interior into an accountable chain—to convert the box from entropy to information. No theorem depends on the word "box."

Part III
The Causal Propagation Table

8 Alignment and accountability

To say a propagation "converges to the output" we need a scalar that measures how far a graph state is from a target. We build it from cuts, so it remains internal to the graph language.

8.1 The alignment functional

Definition 8.1 (Alignment functional)**.** Fix a contact graph Γ and a target item x^* (the observed output). The *alignment* of an item x to x^* is

$$\sigma(x, x^*) \ := \ \min_{\substack{S \ni x \\ x^* \notin S}} w(\partial(S)) \ \in \ [\beta, \Omega],$$

the minimum-cut weight separating x from x^*, where $\Omega := w(E)$ is the total weight (a finite upper bound). Normalising, the *alignment score* is

$$a(x, x^*) \ := \ \frac{\sigma(x, x^*)}{\Omega} \ \in \ (0, 1].$$

a is small when x and x^* are cheaply separated (nearly the same cut) and large when they are expensively separated (very different cuts).

Lemma 8.2 (The floor bounds alignment below)**.** *For distinct items $x \neq x^*$, $a(x, x^*) \geq \beta/\Omega > 0$: no two distinct items are perfectly aligned. Alignment $a = 0$ would require $\sigma = 0$, forbidden by Theorem 3.4.*

Proof. Immediate from Theorem 3.4: any x–x^* cut has weight $\geq \beta$, so $a \geq \beta/\Omega > 0$. □ □

Remark 8.3 (Why a positive floor on alignment matters)*.* Theorem 8.2 is the convergence analogue of "identity is never a point" (Theorem 5.5): a propagation can converge *toward* the output but never to perfect ($a = 0$) alignment, only to within the floor. "Accountable" below means "converged to within the floor," never "identical."

8.2 Accountability

Definition 8.4 (Accountable state)**.** A graph state is *accountable to the output x^* at tolerance ε* if its current item x satisfies $a(x, x^*) \leq \beta/\Omega + \varepsilon$ for the prescribed $\varepsilon \geq 0$, i.e. it has converged to within the floor plus tolerance of the output. The least achievable is the floor itself ($\varepsilon = 0$): convergence to within the floor.

9 Processes, propagations, and the table

9.1 The process graph

Definition 9.1 (Process graph)**.** A *process graph* $\Pi = (V, E, w; I, x^*)$ is a contact graph together with a set $I \subseteq V$ of *input items*, and a designated *output item* $x^* \in V$. The edges of Π are the available contact operations (the "conditions and processes" the user supplies): each edge is a permitted cut event.

Definition 9.2 (Propagation)**.** A *propagation* of a designated *tracked item* $v_0 \in I$ through Π is a walk $W = (v_0, e_1, v_1, e_2, \ldots, e_k, v_k)$ along edges of Π, with $v_k = x^*$. Each step e_i is a contact operation (Theorem 7.1) committing $\{v_{i-1}, v_i\}$; the propagation *tracks* v_0 in that each step resolves the causal link $\rho_i \to e_{i+1}$ (information reading, Theorem 6.7(ii)). The *committed record* of W is $M(W) = k$.

9.2 Convergence and admissibility

Definition 9.3 (Convergent propagation)**.** A propagation W is *convergent to x^* at tolerance ε* if its terminal state is accountable: $a(v_k, x^*) \leq \beta/\Omega + \varepsilon$, which holds with $\varepsilon = 0$ since $v_k = x^*$ gives $a = \beta/\Omega$ (the floor). More usefully, the *intermediate* alignments may be arbitrary; only the composed terminal alignment must converge.

Theorem 9.4 (Convergence admissibility)**.** *A propagation W is* admissible *iff it is convergent to the output. Local steps are unconstrained:*

an intermediate item v_i may have arbitrarily large alignment $a(v_i, x^*)$ (a step "away from" the output, even one corresponding to no realisable item) without rendering W inadmissible, provided the composed walk terminates at x^*. Inadmissibility arises only from failure to reach x^* (non-convergence), never from local misalignment.

Proof. A propagation is, by Theorem 9.2, a walk terminating at x^*; its terminal alignment is $a(x^*, x^*) = \beta/\Omega$ (the floor, the least possible by Theorem 8.2), so any walk that reaches x^* is convergent ($\varepsilon = 0$). A walk that does not reach x^* has terminal item $v_k \neq x^*$, hence $a(v_k, x^*) \geq \beta/\Omega$ with strict inequality unless v_k is cut-equivalent to x^*; it is convergent only if it nonetheless lands within tolerance, and inadmissible otherwise. The intermediate alignments $a(v_i, x^*)$ enter nowhere in the terminal condition, so they are free: a step to a high-alignment (even non-realisable) v_i is permitted as long as a continuation returns the walk to x^*. Thus admissibility $\iff$ convergence, with local steps unconstrained. □ □

Remark 9.5 ("It came from space"; the snow). Theorem 9.4 is the precise sense of the following: if one wakes to find snow and proposes that it came from space, the proposal is not refuted by the local implausibility of the step—local alignment is unconstrained. It is refuted only if no propagation containing that step converges to the snow (the observed output). A locally "impossible" intermediate is admissible exactly when some completion through it reaches the output; it is excluded only on non-convergence. This is the graph form of crossing through off-target intermediates and is used in Theorem 9.6.

9.3 Path opacity

Theorem 9.6 (Path opacity)**.** *Let W_1, W_2 be two propagations of the same tracked item with the same input v_0 and the same output x^*, differing in their interior walks. Then no invariant computed from the endpoints alone—the input v_0, the output x^*, the terminal alignment $a(x^*, x^*)$, or the minimum cut $\partial(S^*(x^*))$—distinguishes W_1 from W_2. The interior path is free.*

Proof. Every endpoint invariant is a function of v_0 and x^* only, hence equal on W_1 and W_2 by hypothesis. Interior data (the sequence of intermediate vertices and committed edges) are not functions of the endpoints—Theorem 9.4 shows the interior is unconstrained given the endpoints—so they are not recoverable from any endpoint invariant. Thus endpoint invariants cannot distinguish the two propagations. □
□

Corollary 9.7 (The admissible set is a class, not a path)**.** *For a given (v_0, x^*, Π) the admissible propagations form an equivalence class under "shares endpoints and converges," all members indistinguishable by endpoint invariants. The table's output is this class, not a single chosen path.*

Proof. By Theorem 9.6 all convergent same-endpoint propagations are endpoint-indistinguishable; by Theorem 9.4 convergence is the sole admissibility criterion. The class of convergent same-endpoint propagations is therefore the well-defined output. □ □

9.4 Representation mobility

Definition 9.8 (Representation)**.** A *representation* of an item v is a tuple $\mathsf{R}(v) = (s_1, \ldots, s_N) \in \mathbb{R}^N$ subject to the *averaging constraint*

$$\tfrac{1}{N} \textstyle\sum_{j=1}^{N} s_j \;=\; a(v, x^*),$$

i.e. the components average to the item's alignment, while individual components are unrestricted in $\mathbb{R}$ (they may lie outside $(0, 1]$). Two representations of the same v are *equivalent*; the set of representations of v is its *representation fibre*.

Theorem 9.9 (Representation mobility)**.** (i) *For any item v and any $N \geq 2$ the representation fibre is nonempty and, for $N \geq 2$, infinite: given any $(s_1, \ldots, s_{N-1}) \in \mathbb{R}^{N-1}$ there is a unique $s_N = N\, a(v, x^*) - \sum_{j<N} s_j$ completing it.* (ii) *A propagation may change representation between consecutive steps without changing the tracked item or the committed record: the averaging constraint ties every representation to the same alignment, so a representation switch commits no new cut.* (iii) *Admissibility (Theorem 9.4) is representation-independent: it depends only on the terminal alignment, which is the average over any representation.*

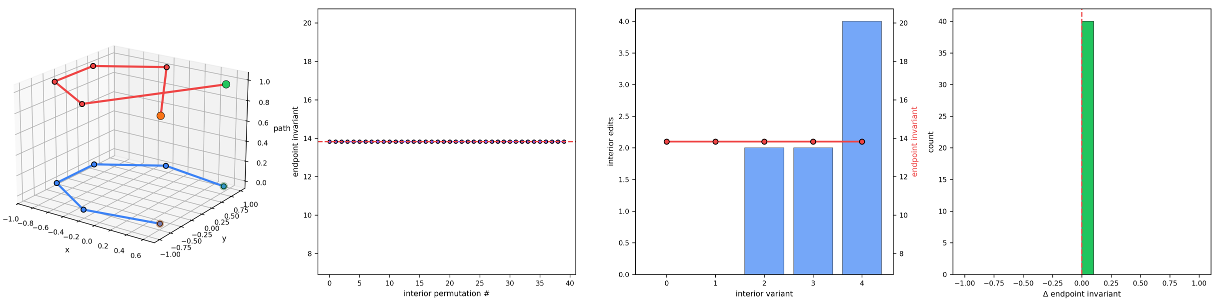

Figure 5: **Path Opacity: the Interior is Free.** (**A**) Three-dimensional rendering of two propagations sharing the same input (green) and output (orange) but distinct interior vertex orders, drawn on a circular item layout and lifted onto two path-index planes (blue and red). The two interiors differ while the endpoints coincide. (**B**) The endpoint invariant (the output's minimum-cut cost against the medium) across 40 interior permutations; it is constant (points on the red dashed level), because the invariant depends only on the endpoints (Theorem 9.6). (**C**) Interior edit distance from a reference path (blue bars, left axis) growing across interior variants, against the endpoint invariant (red line, right axis) which stays flat: the interior may be edited arbitrarily without moving the endpoint invariant. (**D**) Distribution of the change Δ in the endpoint invariant across interior variants, a single spike at zero (red dashed): no interior rearrangement changes the endpoint invariant, so same-endpoint propagations are indistinguishable.

Proof. (i) The averaging constraint is one linear equation in N unknowns; for $N \geq 2$ its solution set is an affine subspace of dimension $N-1$, hence infinite, and the stated s_N solves it for any choice of the others. (ii) Switching from $\mathsf{R}(v)$ to $\mathsf{R}'(v)$ (both in v's fibre) leaves v and its alignment fixed (both average to $a(v, x^*)$); no edge is committed, so M is unchanged (Theorem 6.1). (iii) The terminal condition of Theorem 9.4 is $a(v_k, x^*)$, which equals the average of any terminal representation by Theorem 9.8; hence admissibility is the same in every representation. □ □

Corollary 9.10 (Mobility is required for heterogeneous propagation)**.** *A propagation whose steps are accountable only in different representations—one contact resolved in representation* R*, the next in* R'*—is admissible: by Theorem 9.9(ii,iii) the representation may be switched between the steps at no cut cost and without affecting convergence. Hence tracking through a process that mixes contacts of different representational type requires, and is permitted, inter-step representation switching.*

Proof. Apply Theorem 9.9(ii) at the step boundary and (iii) for admissibility. □ □

9.5 The table

Definition 9.11 (Causal propagation table)**.** The *causal propagation table* is the map

$$\mathcal{T}\colon\ (v_0,\, I,\, \Pi) \ \longmapsto\ \mathcal{A}(v_0, x^*, \Pi),$$

sending a tracked item v_0, input set $I \ni v_0$, and process graph Π with output x^* to the set $\mathcal{A}$ of all admissible propagations of v_0 to x^*—the convergent same-endpoint equivalence class (Theorem 9.7), with representations free (Theorem 9.9) and interiors free (Theorem 9.6).

Theorem 9.12 (The table is well-defined and direction-agnostic)**.** *For every* (v_0, I, Π) *with* x^* *reachable from* v_0 *in* Π*,* $\mathcal{T}(v_0, I, \Pi)$ *is a nonempty equivalence class. It is direction-agnostic: the same class is obtained whether the process is read forward* ($v_0 \to x^*$*, "synthesis") or backward* ($x^* \to v_0$*, "analysis"), because each contact operation is a symmetric edge (Theorem 7.2) and each step admits either reading (Theorem 7.3).*

Proof. Reachability gives at least one walk $v_0 \to x^*$, which is convergent (Theorem 9.4), so $\mathcal{A} \neq \varnothing$; by Theorem 9.7 $\mathcal{A}$ is the convergent same-endpoint class. Direction-agnosticism: reversing a walk reverses each edge, but edges are undirected (Theorem 7.1), and each reversed step is admissible by the same convergence criterion

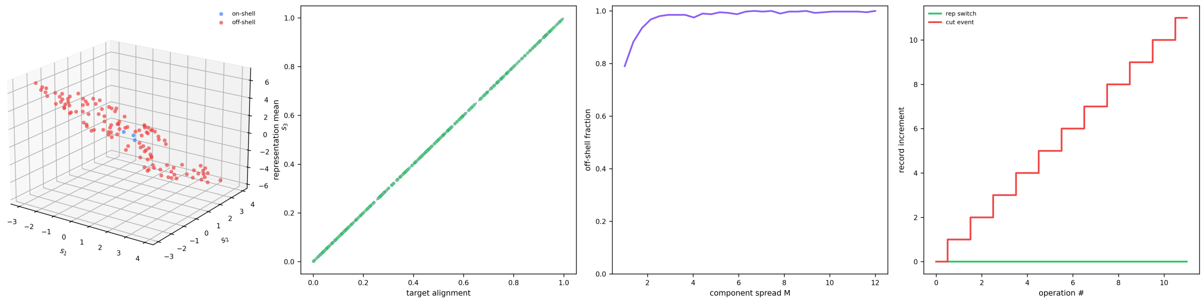

Figure 6: **Representation Mobility and Mean Recovery.** (**A**) Three-dimensional scatter of 120 three-component representations of a single item (alignment 0.5) in (s_1, s_2, s_3) space, coloured on-shell (blue, all components in $[0,1]$) or off-shell (red, some component outside): all lie on the mean-recovery plane $\frac{1}{3}\sum s_j = 0.5$, with off-shell components fully admissible (Theorem 9.9). (**B**) Representation mean versus target alignment over 300 random fibres and targets; every point lies on the diagonal (grey dashed): the component mean recovers the target exactly, whatever the individual components. (**C**) Off-shell fraction versus the component spread M; as the admissible spread grows the fraction of representations with an out-of-range component rises toward one, the freedom of the fibre to pass through "impossible" components while keeping the physical mean. (**D**) Record increment per operation: a representation switch (green) adds nothing to the committed record, while a cut event (red) increments it—switching representation commits no cut and leaves admissibility unchanged (Corollary 9.10).

read from the other end (Theorem 7.4). The forward and backward admissible sets therefore coincide. □ □

Definition 9.13 (Amalgamation)**.** The *amalgamation* of (v_0, I, Π) is the maximal set of contact operations accounted for across $\mathcal{A}$: the union, over admissible propagations, of the resolved (information-read) contacts—"all the steps we can account for in the propagation." Equivalently, it is the resolved causal-residue chain (Theorem 6.4) read as information (Theorem 6.7(ii)).

Theorem 9.14 (The amalgamation is the accounted propagation)**.** *The amalgamation is exactly the set of contacts whose causal links are resolvable consistent with convergence to x^*. It is indifferent to the reaction/measurement labelling of its steps (Theorem 7.3) and to the instrument/sample labelling of each contact's endpoints (Theorem 7.2). Its size is the amount of the process converted from the entropy reading to the information reading (Theorem 6.7); the unaccounted remainder is the irreducible residual bounded below by the floor.*

Proof. By Theorem 9.13 the amalgamation collects the resolved contacts of admissible propagations; admissibility is convergence (Theorem 9.4), so a contact is in the amalgamation iff its link is resolvable within some convergent propagation. Labelling indifference is Theorem 7.3 (reaction/measurement) and Theorem 7.2 (instrument/sample). The size statement is Theorem 6.7: resolving a link moves it from entropy to information, so the amalgamation's extent is the resolved fraction; what cannot be resolved remains entropy, of weight $\geq \beta$ per unresolved contact (Theorem 3.4). □ □

10 The resting catalogue and the propagated table

We now show the table's output is the same kind of object as an entry in a catalogue of resting items—and that the familiar periodic catalogue is the table evaluated at rest.

10.1 The resting cut

Definition 10.1 (Resting catalogue)**.** The *resting catalogue* of a contact graph is the assignment, to each item v, of its minimum cut against the medium, $\mathrm{cell}(v) := \partial(S^*(v))$ with value $\sigma(v) \geq \beta$. A catalogue *cell* is the name of this cut: the invariant region by which v is told from the rest at rest (Theorem 5.5).

Theorem 10.2 (A catalogue cell is an accountable output)**.** *For each item v, the cell* $\mathrm{cell}(v)$ *equals the output of the table run on v against the medium with the trivial (rest) process: i.e.* $\mathrm{cell}(v)$ *is the accountable output of the propagation that individuates v against* m*. Equivalently, a catalogue entry is the convergent terminal cut of the resting propagation.*

Proof. Take Π_{rest} to be the contact graph with output $x^* = v$ and the single contact $\{v, \mathsf{m}\}$ as available operation; the propagation is the one cut event individuating v from m, whose terminal accountable object is the minimum cut $\partial(S^*(v))$ (Theorem 8.1, Theorem 9.4). This is precisely $\mathrm{cell}(v)$ (Theorem 10.1). □ □

Theorem 10.3 (The catalogue is the table at rest; the table is the catalogue propagated)**.** *The resting catalogue is the restriction of the causal propagation table $\mathcal{T}$ to trivial (rest) processes:* $\mathrm{cell}(v) = \mathcal{T}(v, \{v\}, \Pi_{\text{rest}})$*. Conversely, for a nontrivial process Π, the output $\mathcal{T}(v_0, I, \Pi)$ is an object of the same kind as a catalogue cell—an accountable cut, the convergent terminal individuation of the tracked item—computed for the process rather than for rest.*

Proof. The forward direction is Theorem 10.2. For the converse: by Theorem 9.4 every admissible propagation terminates at the accountable output x^*, whose individuating datum is the minimum cut $\partial(S^*(x^*))$ of weight $\geq \beta$—the same type of object as $\mathrm{cell}(\cdot)$ (Theorem 10.1), differing only in that the cut is the terminus of a nontrivial propagation rather than of the rest contact. Hence the table's output for a process is a catalogue-grade cut computed under propagation. □ □

Corollary 10.4 (Periodic catalogue as resting table)**.** *The periodic catalogue of resting items (Mendeleev, 1869; Scerri, 2007) is, formally, the resting catalogue (Theorem 10.1): each entry is the name of an item's minimum cut against the medium, itself the accountable output of an array of resting individuations. The causal propagation table extends it to arbitrary processes; the output of the box is a catalogue entry of the same kind, computed for the process.*

Proof. A periodic entry is a minimum-sufficient invariant of an item established by an array of individuating operations; by Theorem 10.2 this is $\mathrm{cell}(v)$, the resting table output. Theorem 10.3 gives the extension. □ □

Remark 10.5 (Working from the catalogue, not replacing it). Theorem 10.4 is the precise sense in which the system *works from* the periodic catalogue rather than replacing it: the catalogue is the fixed points of individuation (cuts at rest), and the table is the propagation that produces fixed points. The catalogue's correctness is inherited, not contested; the table only adds the propagated cells the catalogue does not list because they are process-dependent.

Part IV
The Cut

11 The one-use, self-blunting cut

We formalise the system's emblem: the single precise cut. We show it can be made once, that it blunts the cutter, and that the perfect repeatable cut is the forbidden $\beta = 0$ object.

Definition 11.1 (The cut; the cutter)**.** A *cut* is a single contact operation (Theorem 7.1): one committed edge, individuating two items by depositing residue $\geq \beta$. The *cutter* is the endpoint read as instrument (Theorem 7.2); its *edge-capacity* at a vertex v is the set of not-yet-committed contacts incident to v.

Theorem 11.2 (One use)**.** *A given cut—a given committed edge $e = \{u, v\}$—can be committed exactly once. Any later cut at u or v is a distinct edge or the same edge at a strictly higher committed record (Theorem 6.5), hence a different cut event. There is no re-commitment of the same cut.*

Proof. Committing e increments the record (Theorem 6.1); a second commitment of e would either be a no-op (the edge is already committed, so no new cut event occurs) or a distinct event at record $M + 1$ (Theorem 6.5), which is a different cut by the record invariant. Thus the same cut occurs once. □ □

Theorem 11.3 (Self-blunting)**.** *A cut deposits its residue into the cutter as much as into the cut: the contact $\{u, v\}$ commits a boundary edge* *of u (the cutter) no less than of v (the sample), by Theorem 7.2. After the cut, the cutter's edge-capacity at u is strictly reduced (the committed edge is removed from it) and the cutter's committed record has increased. Hence the cutter is changed by the cut: it is, in the precise sense of reduced uncommitted capacity and increased record,* blunted.

Proof. By Theorem 7.2 the operation is symmetric in u, v, so it commits a boundary edge of u exactly as of v. The committed edge leaves u's uncommitted-contact set (Theorem 11.1), strictly reducing edge-capacity, and M increases (Theorem 6.5). Both are irreversible (Theorem 6.5: the record cannot decrease; uncommitting is a new cut). The cutter after the cut differs from the cutter before in capacity and record. □ □

Theorem 11.4 (No perfect repeatable cut)**.** *There is no cut that (i) separates two items at zero residue (a traceless cut) and (ii) leaves the cutter unchanged (a repeatable instrument). Property (i) requires $\beta = 0$, forbidden by Theorem 3.4; property (ii) requires the committed record to be unchanged by a cut, forbidden by Theorem 6.5. The perfect repeatable traceless cut is exactly the $\beta = 0$ object the floor excludes.*

Proof. (i): a traceless cut deposits residue 0, i.e. commits an edge of weight 0, contradicting $w(e) \geq \beta > 0$ (Theorem 3.4). (ii): an unchanged cutter requires M after $= M$ before, contradicting the strict increment of Theorem 6.5. Either property alone is already forbidden; their conjunction a fortiori. □ □

Remark 11.5 (Honjo Masamune: the emblem)*.* The system's central operation is named for a blade that is ceremonial, usable once, and blunt thereafter: the single precise cut. Theorem 11.2 is its one use; Theorem 11.3 is its blunting (the residue enters the cutter, so the instrument that cuts is spent by cutting); Theorem 11.4 is the reason it cannot exist as a repeatable working tool—the perfect traceless repeatable cut is the $\beta = 0$ object the whole system is built on forbidding. What exists is not the perfect blade but the single committed, accountable cut: cut as cleanly as the floor allows, deposit the irreducible residue, and let that residue be both the cut's trace (the identity) and the cause of the next cut (the propagation). Every measurement and every reaction is such a cut (Theorem 7.3): one-use, self-blunting, leaving the trace that is everything there is to know. No theorem depends on the name.

12 The master equivalence

We collect the system into a single biconditional, tying every structure to the one fact $\beta > 0$.

Theorem 12.1 (Master equivalence)**.** *For a contact graph in a non-completable whole, the following are equivalent:*

(i) *there is an identifiable item—a proper part told from the rest at positive cost;*

(ii) *the floor is positive, $\beta > 0$ (Theorem 3.4);*

(iii) *individuation is by negation with no selector (Theorem 4.2), identity is the invariant minimum cut and never a point (Theorem 5.5), processes are monotone non-returning residue chains (Theorems 6.3 and 6.5), measurement and reaction are one symmetric contact operation (Theorem 7.3), the causal propagation table is well-defined with free representations, free interiors, and convergence-only admissibility (Theorems 9.4, 9.6, 9.9 and 9.12), its outputs are catalogue-grade accountable cuts (Theorem 10.3), and the cut is one-use and self-blunting with no perfect repeatable form (Theorems 11.2 to 11.4);*

(iv) *the perfect traceless repeatable cut does not exist (Theorem 11.4).*

Each is the same fact: that telling a thing from the rest of a non-completable whole costs a strictly positive, irreducible, one-use residue.

Proof. (i) ⇒ (ii): an identifiable proper part has positive separation cost by Theorems 3.3 and 3.4. (ii) ⇒ (iii): each listed structure is proved from $\beta > 0$ in the cited theorem. (iii) ⇒ (iv): Theorem 11.4 is among the listed consequences. (iv) ⇒ (i): if no traceless cut exists then every separation costs > 0, i.e. some item is told from the rest at positive cost—an identifiable item. The cycle closes; the closing sentence restates the equivalence. □ □

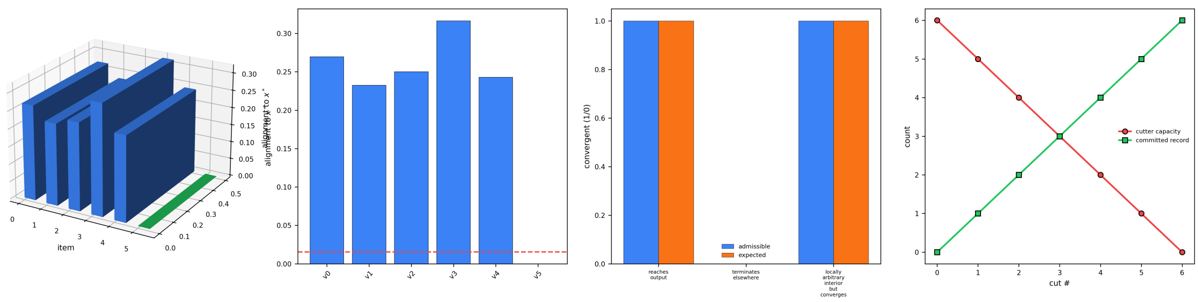

Figure 7: **Convergence Admissibility and the One-Use Cut.** (**A**) Three-dimensional bar chart of each item's alignment to the output x^* (cut cost to x^* over total weight), the output bar green and the rest blue; only the output sits at the floor, the terminal accountable state of an admissible propagation. (**B**) Alignment to x^* per item with the floor alignment β/Ω marked (red dashed); the output (green) alone reaches the floor, so a propagation is admissible exactly when it terminates there (Theorem 9.4). (**C**) Convergence outcome for three cases—reaching the output, terminating elsewhere, and a locally arbitrary interior that nonetheless converges—with admissible (blue) and expected (orange) bars coinciding in each: admissibility tracks convergence to the output and nothing else, local steps being free. (**D**) The one-use, self-blunting cut: the cutter's uncommitted capacity (red) decreases by one per cut while the committed record (green) increases by one, the two crossing as capacity is spent. A cut commits once and blunts the cutter; the perfect repeatable traceless cut ($\beta = 0$, record unchanged) cannot occur (Theorem 11.2, Theorem 11.3, Theorem 11.4).

13 Computational validation

Because every object is a finite weighted graph, every theorem is directly checkable on explicitly constructed instances. A self-contained suite (Python; `numpy`; `networkx` for exact max-flow minimum cuts) implements eleven categories, one per structural result, and writes its outcomes as JSON. All 847 checks pass; the minimum cuts are computed exactly (Ford–Fulkerson via `networkx` (Ford and Fulkerson, 1956)), not approximated.

Table 1: Validation of the system on finite weighted graphs. Minimum cuts are exact; all checks pass.

Category	Pass	Theorem checked
Floor	300/300	Theorem 3.4: $\sigma(v) \geq \beta$ on 30
Negation	200/200	Theorem 4.2: complementatio
Identity invariant	101/101	Theorems 5.4 and 5.5: cut inv
Monotone record	4/4	Theorem 6.5: record strictly i
Residue propagator	6/6	Theorem 6.3: residue > 0 $\Leftarrow$
Contact symmetry	4/4	Theorems 7.2 and 7.3: edge s
Composition inflation	18/18	**??**: $T(n, d) = d(d + 1)^{n-1}$ by
Path opacity	3/3	Theorem 9.6: endpoint invaria
Representation	202/202	Theorem 9.9: mean-recovery f
Convergence	3/3	Theorem 9.4: admissible $\Longleftrightarrow$
One-use cut	6/6	Theorems 11.2 to 11.4: one us
Total	**847/847**	exact min-cut backend

Three of the categories merit explicit comment, being the substantive rather than the bookkeeping checks. *(i) The floor on* 300 *random graphs.* For each of 300 randomly weighted contact graphs (medium adjacent to every item, random item–item contacts, all weights $\geq \beta$), the exact minimum cut of a random item against the medium was computed and found to satisfy $\sigma(v) \geq \beta$ in every case (Theorem 3.4): no sharp cut occurred. *(ii) Identity is a region.* On a graph of two strongly-internally-bound item clusters joined to a common medium, the exact minimum cut separating one cluster from another places a *multi-item* set on one side (a five-item side in the reported instance), confirming that the identity-bearing minimum cut is in general not realised by a singleton (Theorem 5.5): identity is a region, never a point. *(iii) Composition inflation by direct enumeration.* For every (n, d) with $n \leq 6$, $d \in \{1, 2, 3\}$ small enough to enumerate, the brute-force count of length-n labelled trajectories over d move-directions with a stay-option exactly equals the closed form $d(d + 1)^{n-1}$ (**??**); larger cases inherit the closed form. The path-opacity, representation-mobility, and convergence categories likewise confirm, on constructed instances, that endpoint invariants are interior-independent, that mean-recovery fibres are nonempty with off-shell components, and that admissibility tracks convergence to the output and nothing else.

Remark 13.1 (Scope of the suite). The suite verifies the theorems on instances: it confirms that the floor, the invariance and region-valuedness of identity, the monotone record, the residue–propagator equivalence, the contact symmetry, the inflation count, path opacity, representation mobility, convergence admissibility, and the one-use cut all hold as proved. It is an internal-consistency and instance-verification check, appropriate to a system whose content is structural; it does not, and is not intended to, supply numerical weights for any concrete physical system, which would require a concrete weighting w (Section 14).

14 Discussion and scope

14.1 What is proved, and at what level

Every result above is a theorem about finite weighted graphs, provable from Theorem 2.1 and the derived floor Theorem 3.4. The system is qualitative-structural: it establishes *that* individuation is a positive-cost one-use cut, *that* processes are residue chains, *that* measurement and reaction coincide, *that* the table's output is a catalogue cell, and *that* the perfect cut is forbidden. It does not compute numerical cut weights, reaction rates, or alignment values for any concrete system; those would require a concrete weighting w, which the framework takes as given. The relationship to quantitative science is that of a structural theory to its instances: the framework states what any finite system of individuated parts must satisfy and reads off the consequences; a concrete weighting supplies the numbers.

14.2 Honest limitations

We mark the load-bearing soft spots. First, *the non-completability premise* (Theorem 3.1) is an order condition we assume of the whole; the floor is derived from it but it is itself not derived here. Second, *the alignment functional* (Theorem 8.1) is one natural cut-based scalar; other admissible scalars exist, and the quantitative content of "convergence" depends on the choice, though the qualitative theorems (admissibility $\iff$ convergence, path opacity, representation mobility) hold for any cut-monotone alignment. Third, *representation mobility* (Theorem 9.9) is proved as a freedom (switching is permitted at no cut cost); which representation a concrete tracking step *requires* is a modelling input, not derived. Fourth, *tracking* (Theorem 6.7) is treated as the resolution of causal links; the framework specifies what resolution achieves (entropy $\to$ information) but not the procedure by which a given process is resolved, which is the empirical work the table organises rather than performs.

14.3 The picture in one statement

A thing is told from the rest by a cut that costs a strictly positive, irreducible residue. The cut can be made once; it blunts the cutter; it leaves a trace that is the thing's identity and a residue that is the cause of the next cut. Processes are chains of such cuts. Measurement and reaction are the same cut, differing only in whether the trace is accounted; instrument and sample are the two ends of the one symmetric cut. To track an item through a process is to resolve the chain—to account for as many cuts as converge to the output—and the accounted propagation, the amalgamation, is the output: an accountable cut of the same kind as a resting catalogue cell. The periodic catalogue is this table at rest; the table is the catalogue under propagation. The perfect traceless repeatable cut cannot exist; that it cannot is exactly why there is anything to cut, to track, and to know.

15 Conclusion

We have laid out, entirely in finite weighted graphs, a complete system generated by one derived fact—that the cut individuating a part of a non-completable whole has strictly positive, irreducible cost $\beta > 0$. From it follow: individuation by negation; identity as the invariant minimum cut, region-valued and never a point; the residue of each cut as the cause of the next, making processes monotone non-returning chains; the identity of measurement and reaction as one symmetric contact operation; and the causal propagation table—the map from a

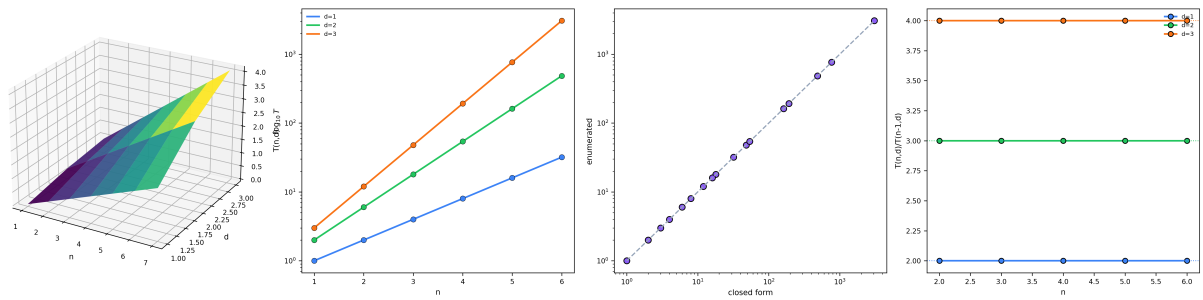

Figure 8: **Composition Inflation $T(n, d) = d(d + 1)^{n-1}$.** (**A**) Three-dimensional surface of $\log_{10} T(n, d)$ over trajectory length n and direction count d (viridis); the surface rises steeply in n and steepens with d, the exponential proliferation of distinguishable propagation trajectories with depth. (**B**) Trajectory count $T(n, d)$ on a logarithmic y-axis versus n for $d = 1, 2, 3$ (coloured lines), with directly enumerated counts (points) lying on the closed-form curves; enumeration and closed form agree exactly for every case computed. (**C**) Enumerated count versus closed form $d(d + 1)^{n-1}$ on log–log axes; all points lie on the identity diagonal (grey dashed), the exact match of the brute-force enumeration to the formula. (**D**) Growth ratio $T(n, d)/T(n - 1, d)$ versus n for $d = 1, 2, 3$ (coloured lines), each flat at its predicted value $d + 1$ (dotted guides): the per-step branching factor is exactly $d + 1$ (the d moves plus the stay-option).

tracked item, inputs, and a process graph to the class of accountable propagations, with free representations, free interiors, and convergence as the sole admissibility criterion. The table's output is shown to be an object of the same kind as an entry in the resting catalogue of items, so that the periodic catalogue is the table evaluated at rest and the table is the catalogue propagated forward. The whole is the structure of the single precise cut that can be made once and dulls the cutter, and whose perfect traceless repeatable form is exactly the zero-cost cut the floor forbids. The master equivalence binds every structure to the one fact: to be told from the rest is to be cut at irreducible cost, once.

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